

DS10 Compilation Instructions

— Environment Configuration

1. Compilation environment

Please visit the following address to download the compilation chain package "owtoolchain.zip"

<https://drive.google.com/file/d/1yVfltWdgUpjrMy-ondYa0T5ihrqVkj3v/view?usp=sharing>

Extract the owtoolchain.zip file from the tool And put it on the C drive (compiler default path, Pay attention to the decompressed path) View the first line of build.bat in the demo

i > Windows-SSD (C:) > owtoolchain

名称	修改日期
ARM_Compiler_5	2024/11/21 16:31
gcc-arm-none-eabi	2024/11/21 16:31
gnumake	2024/11/21 16:31
Perl	2024/11/21 16:32

```
d.bat
set cross_tool=C:\owtoolchain
@echo off
chcp 65001 >nul

set CURDIR=%~dp0
set build_target=%1
set ARMCC_PATH=
set PATH_TMP=
@for %%i in (armcc.exe) do (
    @if not "%~$PATH:i"==" " (
        set PATH_TMP=%~$PATH:i
        SETLOCAL ENABLEDELAYEDEXPANSION
        for /f "delims=" %%i in ('armcc --vsfn') do (set ret=!ret! &
            (echo !ret! | findstr /C:"Software supplied by" >nul) &
        )
        ENDLOCAL
        goto SITE2
    ) else (
        goto SITE2
    )
)
```

2. Open the project directory, enter the demo directory, double-click project_build.bat

名称	修改日期	类型	大小
ql-application	2025/12/19 16:04	文件夹	
ql-config	2025/12/19 16:04	文件夹	
ql-cross-tool	2025/12/19 16:04	文件夹	
app_start_addr.py	2025/9/1 10:30	Python 源文件	8 KB
build.bat	2025/12/16 17:10	Windows 批处理文件	5 KB
config.ini	2025/6/19 18:33	INI 文件	1 KB
devtype.conf	2025/12/19 17:26	CONF 文件	1 KB
make_ota_firmware.bat	2025/12/19 13:01	Windows 批处理文件	2 KB
module.conf	2025/12/19 17:26	CONF 文件	1 KB
project_build.bat	2025/12/22 10:44	Windows 批处理文件	6 KB
project_name.conf	2025/12/19 17:26	CONF 文件	1 KB

After running the script, the following window will pop up

```

main menu
=====

1:build clean
2:build app
3:packet firmware (zip file)
4:packet APP OTA file(bin file,Step 3 must be executed first before generating OTA firmware )
5:set device type
0:quit

prj-dir:D:\git\DS10\demo\
please select item number, then click enter button:

```

- Set the device type and enter the number 5 in the newly popped up window

```

main menu
=====

1:build clean
2:build app
3:packet firmware (zip file)
4:packet APP OTA file(bin file,Step 3 must be executed first before generating OTA firmware )
5:set device type
0:quit

prj-dir:D:\git\DS10\demo\
please select item number, then click enter button:5

```

- Continue pressing the enter key

```

=====PROJECT INFO=====
project name:EC600MEU_LA
rust_ver:
extflash:EXT_FLASH_M08
APP_MODE:XIP
DEM_cfg:
=====
ext flash 8m
XIP APP

*****PRJ_CUR=XIP*****

=====Build Project EC600MEU_LA=====

The system cannot find the path specified.
external

*****ERROR*****
We didn't find any packaging scripts

*****ERROR*****
We didn't find any packaging scripts2

*****ERROR*****
MODULE:EC600MEU_LA *****

```

5. Continue pressing the enter key

```

          main menu
          =====

          1:build clean
          2:build app
          3:packet firmware (zip file)
          4:packet APP OTA file(bin file,Step 3 must be executed first before generating OTA firmware )
          5:set device type
          0:quit

          prj-dir:D:\git\DS10\demo\

please select item number, then click enter button:
          |

```

After entering the menu again, it indicates that the configuration has been completed

二 Compilation project

1. During the first compilation, you can input 1 build clean to avoid leaving any other compiled information
2. Enter 2 to build the app

```

          main menu
          =====

          1:build clean
          2:build app
          3:packet firmware (zip file)
          4:packet APP OTA file(bin file,Step 3 must be executed first before generating OTA firmware )
          5:set device type
          0:quit

          prj-dir:D:\git\DS10\demo\

please select item number, then click enter button:2

```

```

.../extra/widgets/msgbox/lv_msgbox.o build/obj/extra/widgets/span/lv_span.o build/obj/extra/widgets/spinbox/lv_spinbox.o build/obj/extra/widgets/spinner/lv_spinner.o build/obj/extra/widgets/tabview/lv_tabview.o build/obj/extra/widgets/tileview.o build/obj/extra/widgets/win/lv_win.o build/obj/extra/themes/basic/lv_theme_basic.o build/obj/extra/themes/default/lv_theme_default.o build/obj/extra/themes/mono/lv_theme_mono.o
umake[1]: Leaving directory `D:/git/DS10/demo/ql-application/threadx/interface/lvgl'
-----link [build/app.elf]-----
C:\owtoolchain\gcc-arm-none-eabi\bin\arm-none-eabi-ld.exe -gc-sections -TD:\git\DS10\demo\ql-application\threadx\interface\lvgl\build\app.map -Map build/app.map -o build/app.elf --whole-archive D:/git/DS10/demo/ql-application/threadx/evb_audio/build/evb_audio.lib D:/git/DS10/demo/ql-application/threadx/interface/lvgl/build/lvgl.lib common/lib/ql_common_api.lib common/lib/HTTPClient.lib common/lib/mqtt.lib common/lib/serialdev.lib common/lib/cjson/lib/mbedtls.lib common/lib/ym_common.lib common/lib/alipay.lib common/lib/com_zip.lib --no-whole-archive C:\owtoolchain\gcc-arm-none-eabi\lib\thumb\v7+fp\softfp\libc.a C:\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7+fp\softfp\libgcc.a C:\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7+fp\softfp\libm.a C:\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7+fp\softfp\libc.a C:\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7+fp\softfp\libm.a C:\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7+fp\softfp\libnosys.a C:\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\9.2.1\thumb\v7+fp\softfp\libgcc.a
-----objcopy [build/app.bin]-----
C:\owtoolchain\gcc-arm-none-eabi\bin\arm-none-eabi-objcopy.exe -O binary build/app.elf build/app.bin
D:/git/DS10/demo/ql-application/threadx/../../ql-cross-tool/win32/host/tools/crc_bin.exe build/app.bin
-----CC505C-----
CC Compiling Finished Sucessfully.
The target image is in the 'build' directory.
-----
App compilation successfully.Please confirm the compilation process prompt.

```

The above information indicates that the compilation was successful

3. Pack the firmware, press Enter to return to the menu, and enter 3 packet firmware

```

main menu
=====
1:build clean
2:build app
3:packet firmware (zip file)
4:packet APP OTA file(bin file,Step 3 must be executed first before generating OTA firmware )
5:set device type
0:quit

prj-dir:D:\git\DS10\demo\
please select item number, then click enter button:3

```

```

generating target release package ...
Done.
Release package generated successfully!
1 file(s) copied.
1, 3289088, 3289088, 0, 1, 2, 198CEB59013FA7CAE26885B1696EA4A6
2, 126313, 126313, 0, 1, 2, 85E0DDC1C1FFD1065FC752A53ED9B015
3, 1998, 1998, 0, 1, 2, 079F37F839464DE9B15A4F3BBDCBDB84
3C023E35ECE19B91B493D51ABADDCD82
1 file(s) copied.
1 file(s) copied.
1 file(s) copied.
1 file(s) copied.
1 file(s) copied.
1 file(s) copied.
"build firmware without dbg compress"
=====SUCCESS=====
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** ** ** ** ** 
** ** ** ** ** ** ** 
Firmware packet successfully◆◆Please confirm the compilation process prompt.

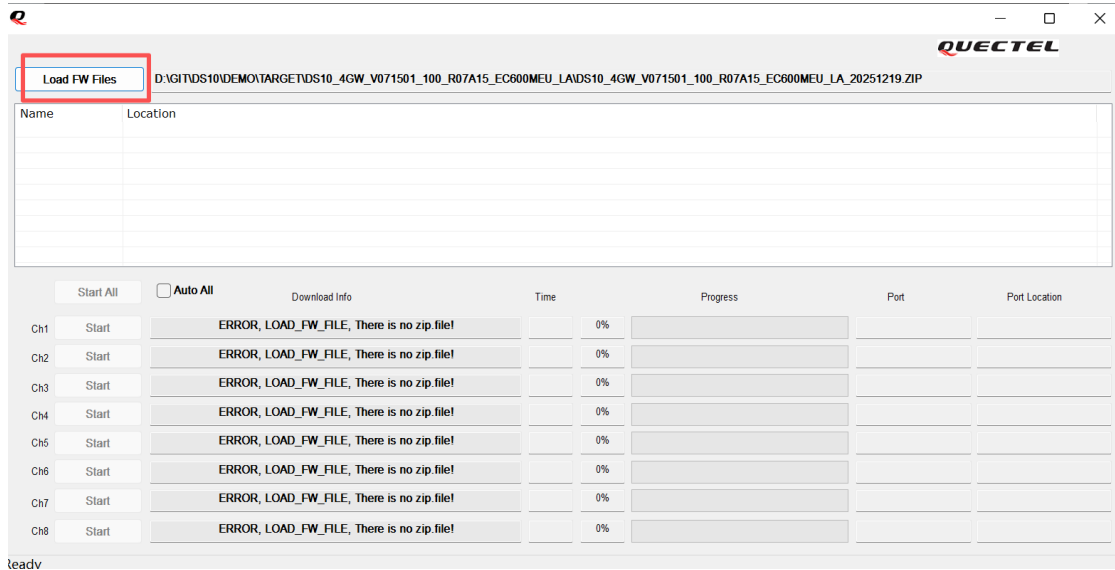
```

The above information indicates that the firmware has been successfully packaged.

The firmware file will be available. Generate a zip format firmware package under demo/target/DS10_4GW-V071501_100_S07A15_SC600MEU_LA

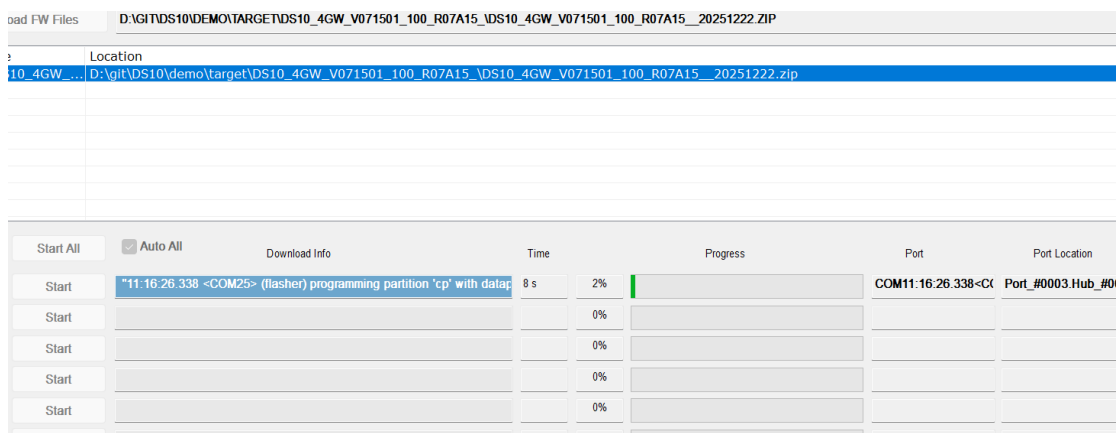
≡ Firmware Download

1. Run tool/download_tool/QMulti_SL_V2.6/QMulti_SL_V2.6.exe

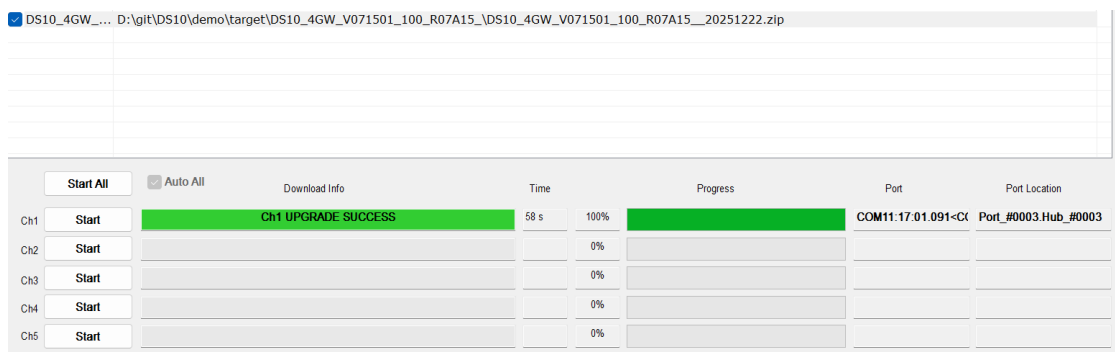


2. Choose to generate a zip file

When the device is turned off, insert the USB cable and press the power button. The tool will detect the device status and start firmware download



3. firmware download success



4. Close the download tool and unplug the USB first, then press the power button to turn on the device, and the new firmware will run (Please note that the download tool

must be closed, otherwise the device will enter download mode again)

四 Generate OTA firmware

```
main menu
=====

1:build clean
2:build app
3:packet firmware (zip file)
4:packet APP OTA file(bin file)
7:set device type
0:quit

    nts: if you select item 6 and item 7, kernel will be rebuilt , dn't do it if no needs. just only build app and
    packet firmware in normol

    prj-dir:D:\git\DS10\demo\

please select item number, then click enter button:
```

You need to follow steps 2, 3, and 4 in sequence.

```
ProcessBLF return....

Writing Device Header:
m_dev.nOfImages = 1
m_dev.DeviceFlags = 0

Image file name:D:\git\DS10\demo\ql-cross-tool\FBFMake_CF\full\customer_app.bin
image header:
image->Image_ID = 0x00000030
image->Image_In_TIM = 0x0
image->Flash_partition = 0x00000000
image->Flash_erase_size = 0x000a2300
image->commands = 0x3
image->First_Sector = 0x1
image->length = 0x000a2300
image->Flash_Start_Address = 0x00434000
image->ChecksumFormatVersion2 = 0xa22f8

dump to fbf file
Finish to dump to fbf file
nOfImages = 1

FBFMake for WEB successfully...
移动了      1 个文件。
OTA Firmware create success
finish
    OTA bin packet successfully, Please confirm the compilation process prompt.
```

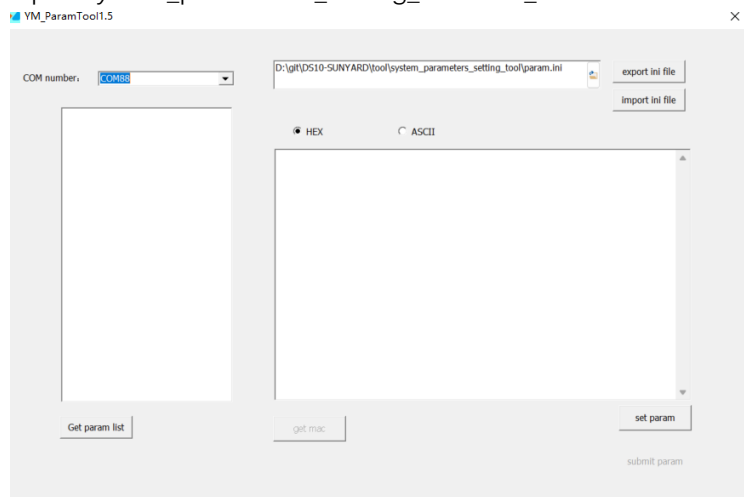
After the fourth step is successfully executed, OTA firmware will be generated in the current directory of the script It is a bin file.

 .vscode	20
 ql-application	20
 ql-config	20
 ql-cross-tool	20
 target	20
 4GW_V071501_100_R07A15_OTA.bin	20
 app_start_addr.py	20
 build.bat	20
 config.ini	20
 devtype.conf	20
 make_ota_firmware.bat	20
 module.conf	20
 project_build.bat	20
 project_name.conf	20

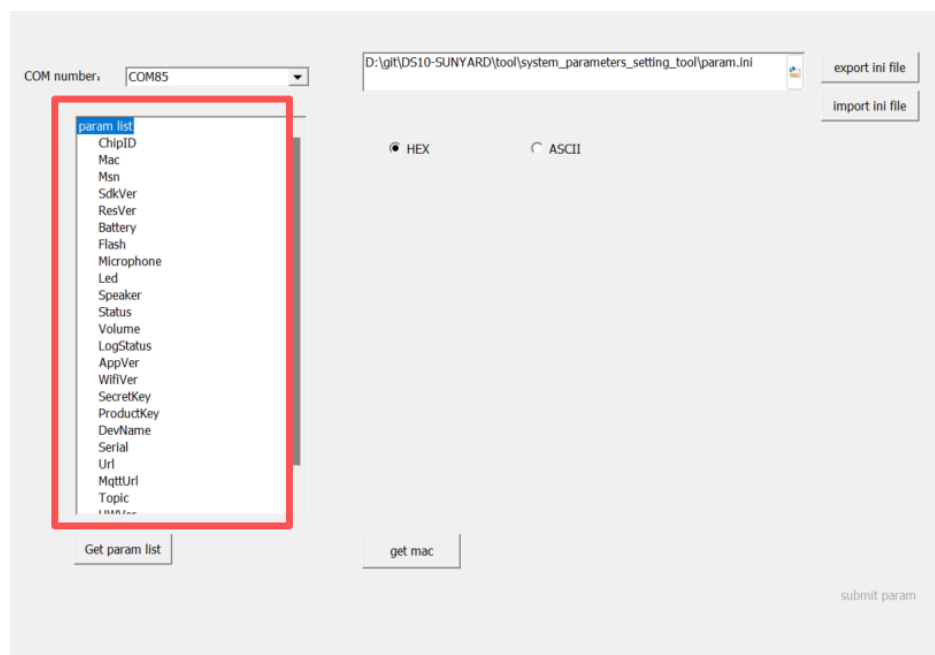
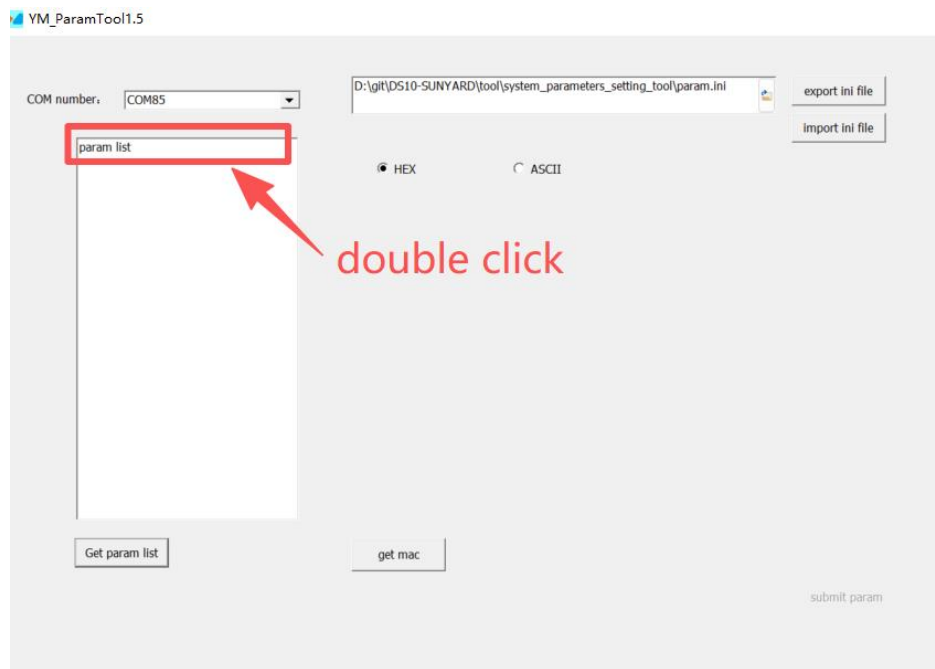
OTA firmware can be updated through TMS.

五 Debug Mode

1. Install the usb driver. tool/driver/Quectel_Windows_USB_Driver(A)_Customer_V1.0.15.zip
2. When the device is turned off, press and hold the function key and power key simultaneously to turn it on
3. When hearing the broadcast 'enter factory mode', the device enters the setup mode
4. Open system_parameters_setting_tool/ YM_ParamTool1.5.exe



5. The device will list 4 serial ports and select the first one Click the 'Get param list' button



6. Click on the LogStatus parameter

COM number: COM85

D:\git\DS10-SUNYARD\tool\system_parameters_setting_tool\param.ini

export ini file

import ini file

param list

- ChipID
- Mac
- Msn
- SdkVer
- ResVer
- Battery
- Flash
- Microphone
- Led
- Speaker
- Status
- Volume
- LogStatus
- AppVer
- WifiVer
- SecretKey
- ProductKey
- DevName
- Serial
- Url
- MqttUrl
- Topic

Get param list

get mac

set param

submit param

00

HEX ASCII input length:1 max length:1

“00” Indicates release status, unable to retrieve logs Need to be changed to “02”

COM number: COM85

D:\git\DS10-SUNYARD\tool\system_parameters_setting_tool\param.ini

export ini file

import ini file

param list

- ChipID
- Mac
- Msn
- SdkVer
- ResVer
- Battery
- Flash
- Microphone
- Led
- Speaker
- Status
- Volume
- LogStatus
- AppVer
- WifiVer
- SecretKey
- ProductKey
- DevName
- Serial
- Url
- MqttUrl
- Topic

Get param list

get mac

set param

submit param

02

HEX ASCII input length:1 max length:1

COM number: COM85

D:\git\DS10-SUNYARD\tool\system_parameters_setting_tool\param.ini

export ini file

import ini file

param list

Get param list

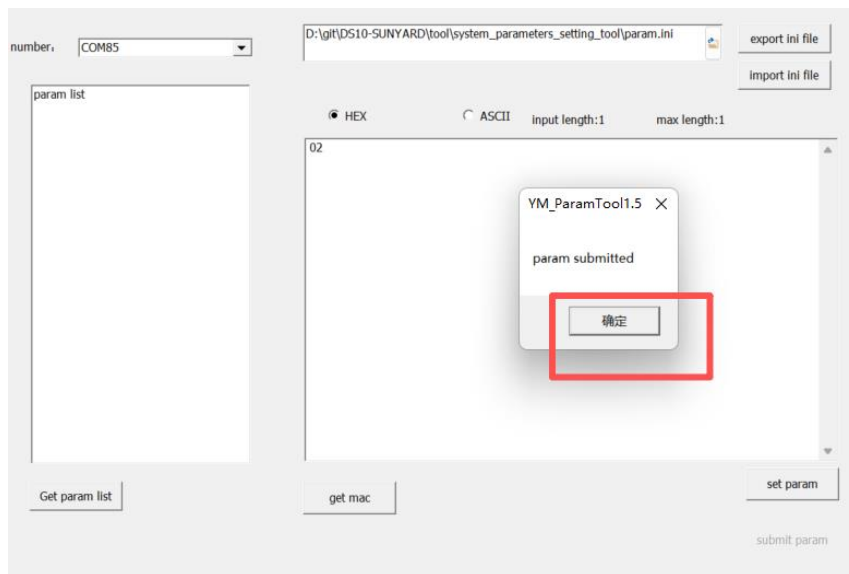
get mac

set param

submit param

02

HEX ASCII input length:1 max length:1

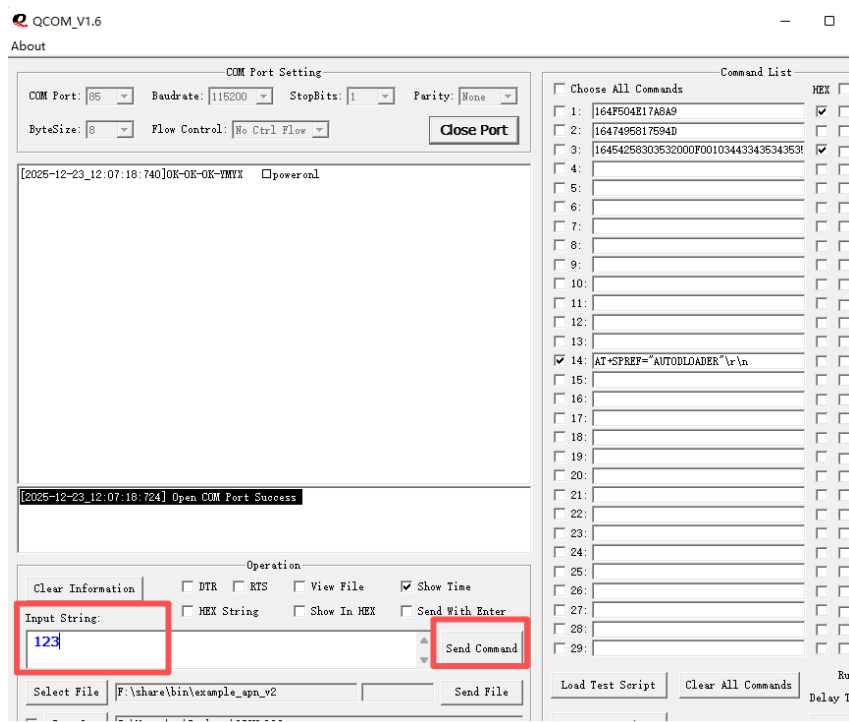


After clicking confirm, the setting was successful

7. After clicking confirm, the setting was successful

After the device is turned on, it will first broadcast "welcome to cloud speaker", followed by a beep sound.

Open a serial port tool and select the first serial port for enumeration (along with the serial port for setting parameters)



Enter any string in the sending area and click send

COM Port: 85Baudrate: 115200StopBits: 1Parity: NoneByteSize: 8Flow Control: No Ctrl FlowOpen Port

[2025-12-23_12:07:07:18:724] Open CUM Port Success

Clear Information☐ DTR☐ RTS☐ View File☒ Show Time☐ HEX String☐ Show In HEX☐ Send With Enter

Input String: 123Send Command

Select File: F:\share\bin\example_app_v2Send File

Command List

☐ Choose All CommandsHEX

1: 164F504E17A8A9☒

2: 1647495817594D☐

3: 1645425830353200F00103443343534353☒

4: ☐

5: ☐

6: ☐

7: ☐

8: ☐

9: ☐

10: ☐

11: ☐

12: ☐

13: ☐

14: AT+SFREQ="AUTODLOADER"\r\n☒

15: ☐

16: ☐

17: ☐

18: ☐

19: ☐

20: ☐

21: ☐

22: ☐

23: ☐

24: ☐

25: ☐

26: ☐

27: ☐

28: ☐

29: ☐

Load Test ScriptClear All CommandsRun

The serial port tool has log output